

Bit Manipulation (Assignment Solutions)

Question 1 :

```
void clearBitsInRange(int num, int i, int j) {  
    int a = (~0) << (j+1);  
    int b = (1 << i) - 1;  
  
    int mask = a | b;  
    num = num & mask;  
  
    cout << num << endl;  
}
```

Question 2 :

```
int singleNumber(vector<int>& nums) {  
    int answer = 0  
    for(int i=0; i<nums.size(); i++) {  
        answer ^= nums[i];  
    }  
  
    return answer;  
}
```

Question 3 :

```
int xorBeauty(vector<int>& nums) {  
    int a = 0, b = 0;  
    for(int i=0; i<nums.size(); i++){  
        a = a | nums[i];  
        b = b ^ nums[i];  
    }  
    return ( a & b );  
}
```